



## **FATE MONTHLY REPORT**

Doc.No : FATE-001  
Version : 1.0  
Date : 11-10-2023



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# **FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE**

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Performance assessment  
October 2023



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## 1 Title for Night 1

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## 1.1 Title for Figures of Night 1

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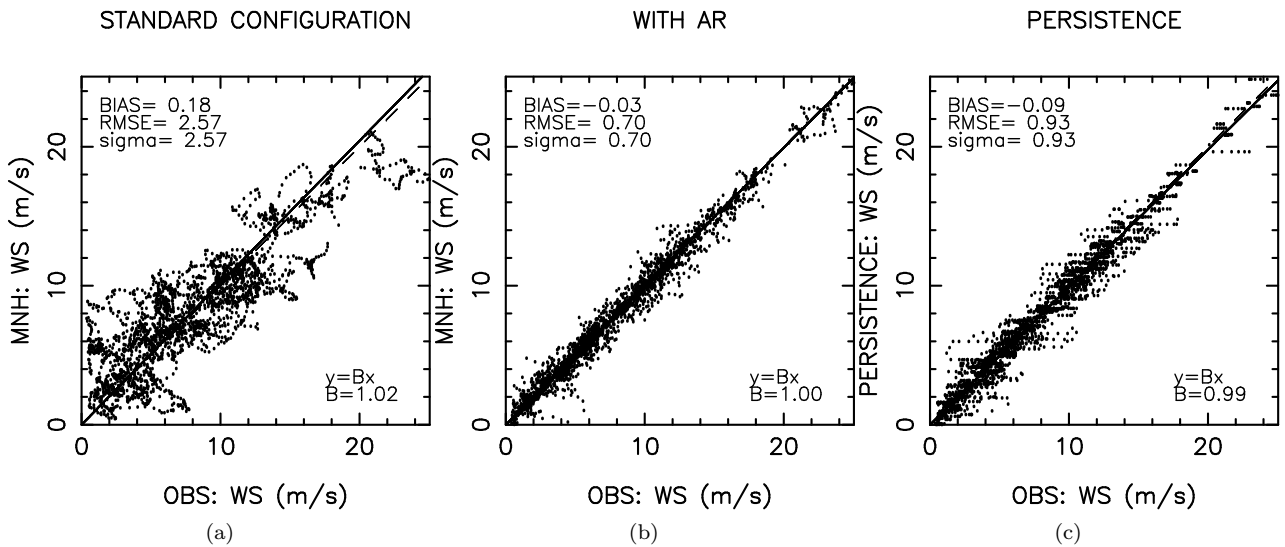


Figure 1: wind speed ( $ms^{-1}$ ): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

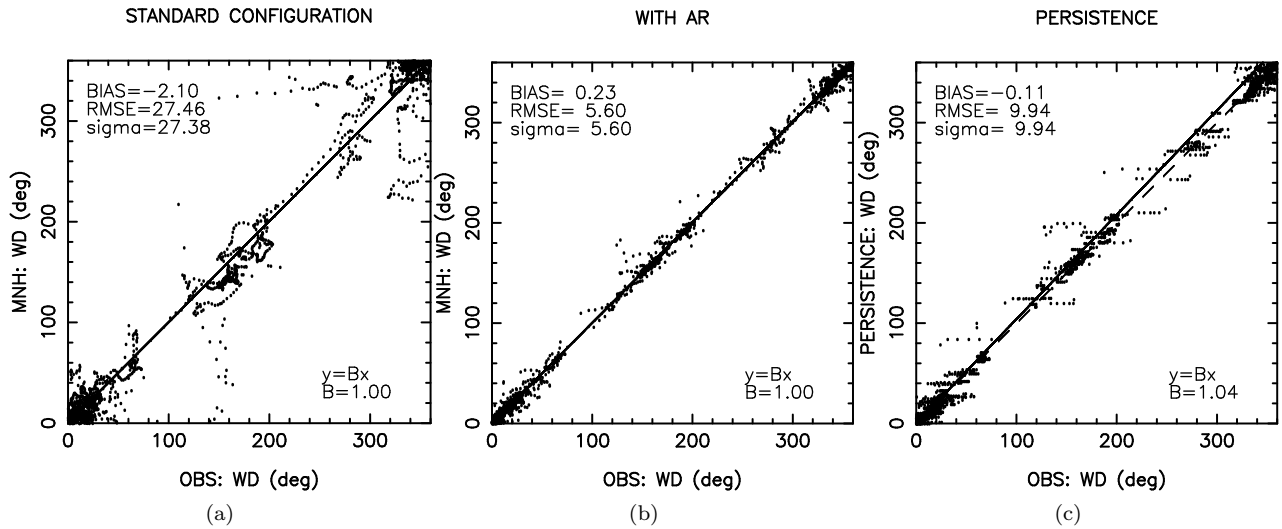


Figure 2: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

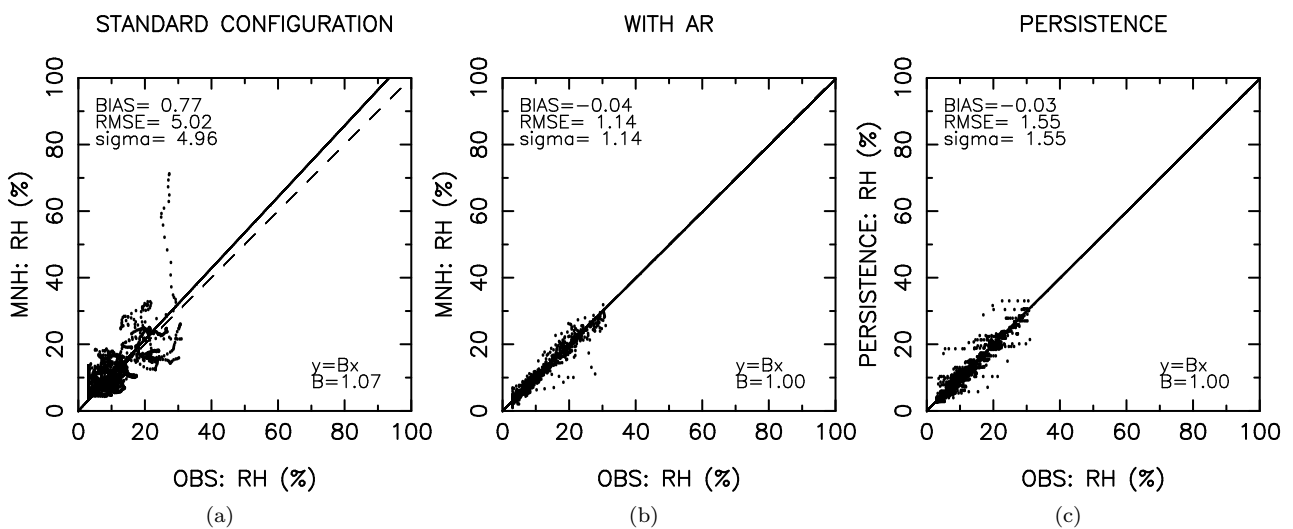


Figure 3: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

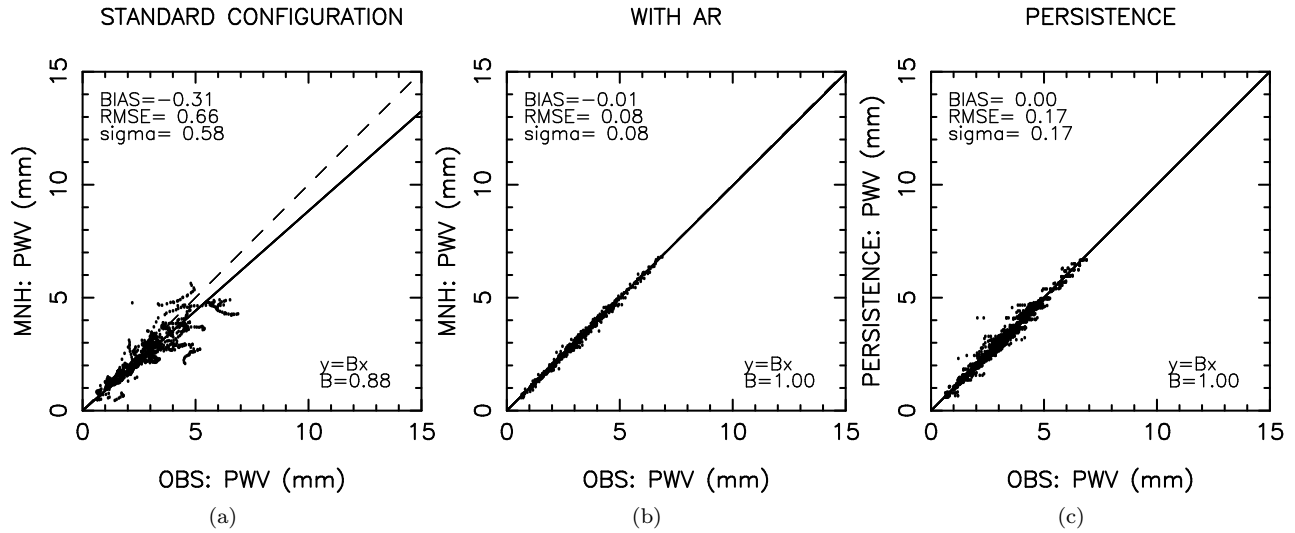


Figure 4: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

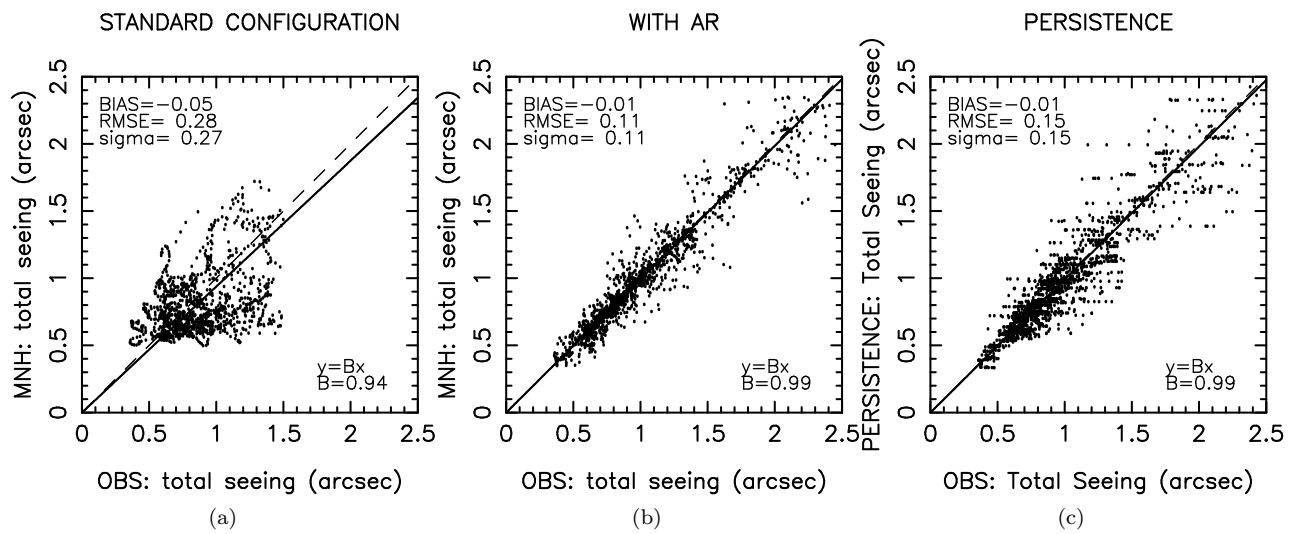


Figure 5: total seeing (arcsec): (a) STANDARD CONFIGURATION ( $< 1.5''$ ), (b) WITH AR (1H), (c) PERSISTENCE (1H).

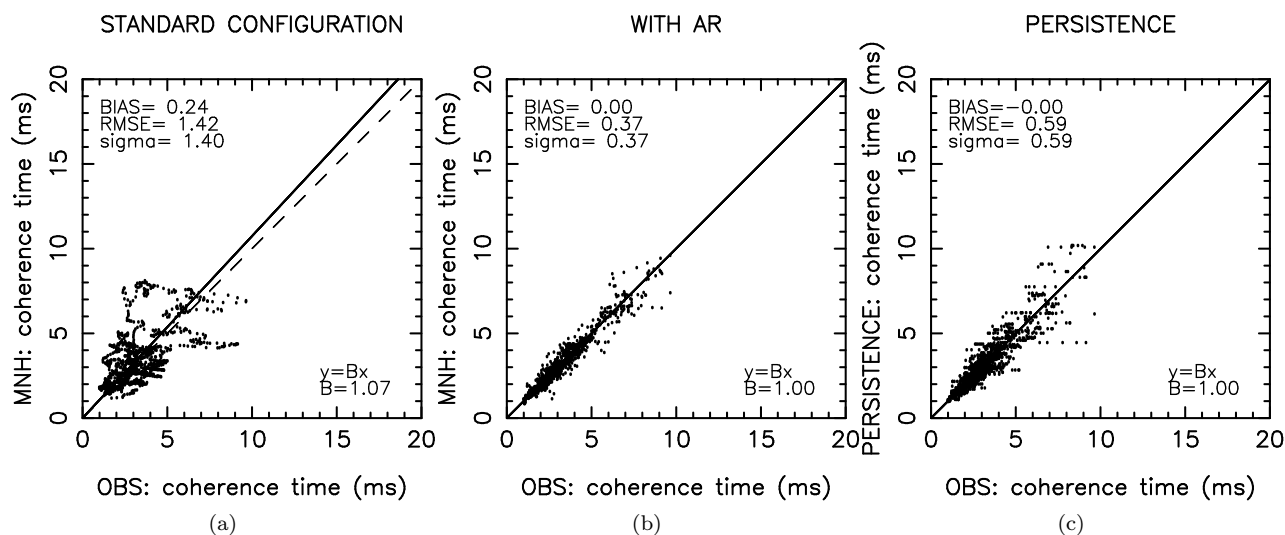


Figure 6: coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

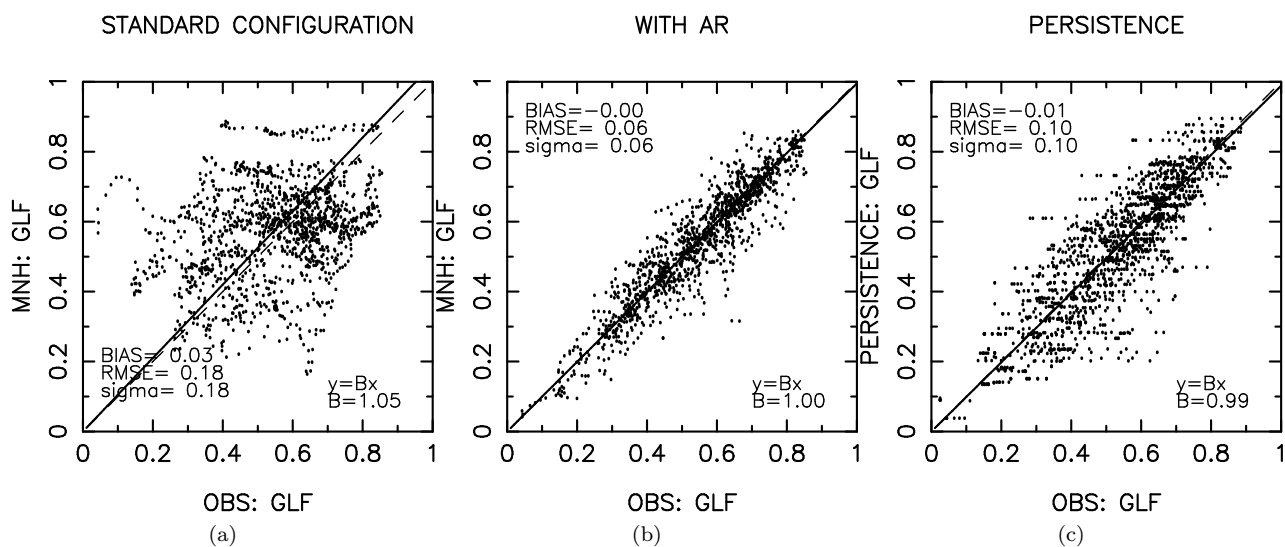


Figure 7: ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

## 1.2 Title for Statistics of Night 1

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### 1.3 Title for Contingency tables of Night 1

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| wind speed<br>$ms^{-1}$                                                   | Observations    |                 |           |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                           | ws < 5.3        | 5.3 < ws < 10.1 | ws > 10.1 |
| Model data                                                                | ws < 5.3        | 501             | 80        |
|                                                                           | 5.3 < ws < 10.1 | 319             | 617       |
|                                                                           | ws > 10.1       | 6               | 129       |
| Sample size: 3996; PC=69.8%; EBD=0.2%; POD1=60.7%; POD2=74.7%; POD3=74.1% |                 |                 |           |

Table 1: Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind speed<br>$ms^{-1}$                                                   | Observations    |                 |           |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                           | ws < 5.3        | 5.3 < ws < 10.1 | ws > 10.1 |
| Model data                                                                | ws < 5.3        | 778             | 45        |
|                                                                           | 5.3 < ws < 10.1 | 48              | 743       |
|                                                                           | ws > 10.1       | 0               | 38        |
| Sample size: 3996; PC=92.7%; EBD=0.0%; POD1=94.2%; POD2=90.0%; POD3=93.9% |                 |                 |           |

Table 2: Contingency table for variable wind speed ( $ms^{-1}$ ) processed with AR (1H)

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| wind direction<br>degree |       | North | East | South | Observations | West |
|--------------------------|-------|-------|------|-------|--------------|------|
| Model data               | North | 1371  | 35   | 14    |              | 45   |
|                          | East  | 25    | 90   | 34    |              | 0    |
|                          | South | 5     | 15   | 345   |              | 0    |
|                          | West  | 45    | 2    | 1     |              | 60   |

Sample size: 3996; PC=89.4%; EBD=1.0%; POD1=94.8%; POD2=63.4%; POD3=87.6%; POD4=57.1%

Table 3: Contingency table for wind direction (degree) in standard configuration

| wind direction<br>degree |       | North | East | South | Observations | West |
|--------------------------|-------|-------|------|-------|--------------|------|
| Model data               | North | 1437  | 17   | 0     |              | 3    |
|                          | East  | 5     | 116  | 3     |              | 0    |
|                          | South | 0     | 9    | 389   |              | 0    |
|                          | West  | 4     | 0    | 2     |              | 102  |

Sample size: 3996; PC=97.9%; EBD=0.0%; POD1=99.4%; POD2=81.7%; POD3=98.7%; POD4=97.1%

Table 4: Contingency table for wind direction (degree) in standard configuration

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| relative humidity<br>percent                                              |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | rh < 7.9     | 7.9 < rh < 11.2 | rh > 11.2 |
| Model data                                                                | rh < 7.9        | 466          | 286             | 49        |
|                                                                           | 7.9 < rh < 11.2 | 198          | 330             | 171       |
|                                                                           | rh > 11.2       | 148          | 196             | 592       |
| Sample size: 3996; PC=57.0%; EBD=8.1%; POD1=57.4%; POD2=40.6%; POD3=72.9% |                 |              |                 |           |

Table 5: Contingency table for variable relative humidity (percent) in standard configuration

| relative humidity<br>percent                                              |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | rh < 7.9     | 7.9 < rh < 11.2 | rh > 11.2 |
| Model data                                                                | rh < 7.9        | 774          | 40              | 2         |
|                                                                           | 7.9 < rh < 11.2 | 38           | 728             | 43        |
|                                                                           | rh > 11.2       | 0            | 44              | 767       |
| Sample size: 3996; PC=93.1%; EBD=0.1%; POD1=95.3%; POD2=89.7%; POD3=94.5% |                 |              |                 |           |

Table 6: Contingency table for variable relative humidity (percent) processed with AR (1H)

| precipitable water vapor<br>mm                                            |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | pwv < 1.9    | 1.9 < pwv < 3.1 | pwv > 3.1 |
| Model data                                                                | pwv < 1.9       | 379          | 66              | 0         |
|                                                                           | 1.9 < pwv < 3.1 | 10           | 306             | 141       |
|                                                                           | pwv > 3.1       | 0            | 17              | 248       |
| Sample size: 1998; PC=79.9%; EBD=0.0%; POD1=97.4%; POD2=78.7%; POD3=63.8% |                 |              |                 |           |

Table 7: Contingency table for variable precipitable water vapor (mm) in standard configuration

| precipitable water vapor<br>mm                                            |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | pwv < 1.9    | 1.9 < pwv < 3.1 | pwv > 3.1 |
| Model data                                                                | pwv < 1.9       | 386          | 11              | 0         |
|                                                                           | 1.9 < pwv < 3.1 | 3            | 371             | 14        |
|                                                                           | pwv > 3.1       | 0            | 7               | 375       |
| Sample size: 1998; PC=97.0%; EBD=0.0%; POD1=99.2%; POD2=95.4%; POD3=96.4% |                 |              |                 |           |

Table 8: Contingency table for variable precipitable water vapor (mm) processed with AR (1H)

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| total seeing<br>arcsec                                                     |                 | Observations |                 |           |
|----------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                            |                 | see < 0.7    | 0.7 < see < 1.0 | see > 1.0 |
| Model data                                                                 | see < 0.7       | 221          | 203             | 98        |
|                                                                            | 0.7 < see < 1.0 | 137          | 141             | 151       |
|                                                                            | see > 1.0       | 42           | 56              | 151       |
| Sample size: 3024; PC=42.8%; EBD=11.7%; POD1=55.2%; POD2=35.2%; POD3=37.8% |                 |              |                 |           |

Table 9: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.0

| total seeing<br>arcsec                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | see < 0.7    | 0.7 < see < 1.0 | see > 1.0 |
| Model data                                                                | see < 0.7       | 425          | 44              | 1         |
|                                                                           | 0.7 < see < 1.0 | 35           | 374             | 37        |
|                                                                           | see > 1.0       | 1            | 43              | 422       |
| Sample size: 3024; PC=88.4%; EBD=0.1%; POD1=92.2%; POD2=81.1%; POD3=91.7% |                 |              |                 |           |

Table 10: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

| total seeing<br>arcsec                                                     |                 | Observations |                 |           |
|----------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                            |                 | see < 0.7    | 0.7 < see < 1.0 | see > 1.0 |
| Model data                                                                 | see < 0.7       | 362          | 0               | 92        |
|                                                                            | 0.7 < see < 1.0 | 0            | 391             | 92        |
|                                                                            | see > 1.0       | 38           | 9               | 308       |
| Sample size: 3024; PC=88.4%; EBD=10.8%; POD1=90.5%; POD2=97.8%; POD3=62.6% |                 |              |                 |           |

Table 11: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.24

| total seeing<br>arcsec                                                    |                 | Observations |                 |           |
|---------------------------------------------------------------------------|-----------------|--------------|-----------------|-----------|
|                                                                           |                 | see < 0.7    | 0.7 < see < 1.0 | see > 1.0 |
| Model data                                                                | see < 0.7       | 459          | 0               | 3         |
|                                                                           | 0.7 < see < 1.0 | 1            | 456             | 3         |
|                                                                           | see > 1.0       | 1            | 5               | 457       |
| Sample size: 3024; PC=99.3%; EBD=0.3%; POD1=99.6%; POD2=98.9%; POD3=98.7% |                 |              |                 |           |

Table 12: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

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| coherence time |                    | Observations |                    |              |
|----------------|--------------------|--------------|--------------------|--------------|
| ms             |                    | $\tau < 2.4$ | $2.4 < \tau < 3.5$ | $\tau > 3.5$ |
| Model data     | $\tau < 2.4$       | 266          | 88                 | 12           |
|                | $2.4 < \tau < 3.5$ | 94           | 185                | 132          |
|                | $\tau > 3.5$       | 66           | 153                | 282          |

Sample size: 3024; PC=57.4%; EBD=6.1%; POD1=62.4%; POD2=43.4%; POD3=66.2%

Table 13: Contingency table for variable coherence time (ms) in standard configuration and accuracy 0.0

| coherence time |                    | Observations |                    |              |
|----------------|--------------------|--------------|--------------------|--------------|
| ms             |                    | $\tau < 2.4$ | $2.4 < \tau < 3.5$ | $\tau > 3.5$ |
| Model data     | $\tau < 2.4$       | 392          | 28                 | 0            |
|                | $2.4 < \tau < 3.5$ | 33           | 375                | 28           |
|                | $\tau > 3.5$       | 1            | 23                 | 398          |

Sample size: 3024; PC=91.2%; EBD=0.1%; POD1=92.0%; POD2=88.0%; POD3=93.4%

Table 14: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

| coherence time |                    | Observations |                    |              |
|----------------|--------------------|--------------|--------------------|--------------|
| ms             |                    | $\tau < 2.4$ | $2.4 < \tau < 3.5$ | $\tau > 3.5$ |
| Model data     | $\tau < 2.4$       | 370          | 0                  | 12           |
|                | $2.4 < \tau < 3.5$ | 0            | 378                | 12           |
|                | $\tau > 3.5$       | 56           | 48                 | 414          |

Sample size: 3024; PC=90.9%; EBD=5.3%; POD1=86.9%; POD2=88.7%; POD3=94.5%

Table 15: Contingency table for variable coherence time (ms) in standard configuration and accuracy 1.22

| coherence time |                    | Observations |                    |              |
|----------------|--------------------|--------------|--------------------|--------------|
| ms             |                    | $\tau < 2.4$ | $2.4 < \tau < 3.5$ | $\tau > 3.5$ |
| Model data     | $\tau < 2.4$       | 425          | 0                  | 0            |
|                | $2.4 < \tau < 3.5$ | 0            | 426                | 0            |
|                | $\tau > 3.5$       | 1            | 0                  | 426          |

Sample size: 3024; PC=99.9%; EBD=0.1%; POD1=99.8%; POD2=100.0%; POD3=100.0%

Table 16: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

| ground layer fraction<br>dimensionless | Observations    |                 |           |
|----------------------------------------|-----------------|-----------------|-----------|
|                                        | glf < 0.5       | 0.5 < glf < 0.6 | glf > 0.6 |
| Model data                             | glf < 0.5       | 182             | 76        |
|                                        | 0.5 < glf < 0.6 | 157             | 209       |
|                                        | glf > 0.6       | 92              | 146       |

Sample size: 3024; PC=41.8%; EBD=10.8%; POD1=42.2%; POD2=48.5%; POD3=34.6%

Table 17: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

| ground layer fraction<br>dimensionless | Observations    |                 |           |
|----------------------------------------|-----------------|-----------------|-----------|
|                                        | glf < 0.5       | 0.5 < glf < 0.6 | glf > 0.6 |
| Model data                             | glf < 0.5       | 375             | 55        |
|                                        | 0.5 < glf < 0.6 | 55              | 309       |
|                                        | glf > 0.6       | 1               | 67        |

Sample size: 3024; PC=80.7%; EBD=0.3%; POD1=87.0%; POD2=71.7%; POD3=83.3%

Table 18: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

| ground layer fraction<br>dimensionless | Observations    |                 |           |
|----------------------------------------|-----------------|-----------------|-----------|
|                                        | glf < 0.5       | 0.5 < glf < 0.6 | glf > 0.6 |
| Model data                             | glf < 0.5       | 319             | 17        |
|                                        | 0.5 < glf < 0.6 | 20              | 393       |
|                                        | glf > 0.6       | 92              | 21        |

Sample size: 3024; PC=83.2%; EBD=12.3%; POD1=74.0%; POD2=91.2%; POD3=75.1%

Table 19: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

| ground layer fraction<br>dimensionless | Observations    |                 |           |
|----------------------------------------|-----------------|-----------------|-----------|
|                                        | glf < 0.5       | 0.5 < glf < 0.6 | glf > 0.6 |
| Model data                             | glf < 0.5       | 427             | 3         |
|                                        | 0.5 < glf < 0.6 | 3               | 427       |
|                                        | glf > 0.6       | 1               | 1         |

Sample size: 3024; PC=99.0%; EBD=0.5%; POD1=99.1%; POD2=99.1%; POD3=98.2%

Table 20: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

#### 1.4 Title for PODs tables of Night 1

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| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 60.7     | 94.2         |
| POD2      | 74.7     | 90.0         |
| POD3      | 74.1     | 93.9         |
| PC        | 69.8     | 92.7         |
| EBD       | 0.2      | 0.0          |

Table 21: PODs for wind speed ( $ms^{-1}$ )

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 94.8     | 99.4         |
| POD2      | 63.4     | 81.7         |
| POD3      | 87.6     | 98.7         |
| POD4      | 57.1     | 97.1         |
| PC        | 89.4     | 97.9         |
| EBD       | 1.0      | 0.0          |

Table 22: PODs for wind direction (degree)

**FATE - Monthly Report**

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 57.4     | 95.3         |
| POD2      | 40.6     | 89.7         |
| POD3      | 72.9     | 94.5         |
| PC        | 57.0     | 93.1         |
| EBD       | 8.1      | 0.1          |

Table 23: PODs for relative humidity (percent)

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 97.4     | 99.2         |
| POD2      | 78.7     | 95.4         |
| POD3      | 63.8     | 96.4         |
| PC        | 79.9     | 97.0         |
| EBD       | 0.0      | 0.0          |

Table 24: PODs for precipitable water vapor (mm)

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 55.2     | 92.2         |
| POD2      | 35.2     | 81.1         |
| POD3      | 37.8     | 91.7         |
| PC        | 42.8     | 88.4         |
| EBD       | 11.7     | 0.1          |

Table 25: PODs for total seeing (arcsec) and accuray 0.0

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 90.5     | 99.6         |
| POD2      | 97.8     | 98.9         |
| POD3      | 62.6     | 98.7         |
| PC        | 88.4     | 99.3         |
| EBD       | 10.8     | 0.3          |

Table 26: PODs for total seeing (arcsec) and accuray 0.24

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 62.4     | 92.0         |
| POD2      | 43.4     | 88.0         |
| POD3      | 66.2     | 93.4         |
| PC        | 57.4     | 91.2         |
| EBD       | 6.1      | 0.1          |

Table 27: PODs for coherence time (ms) and accuray 0.0

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| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 86.9     | 99.8         |
| POD2      | 88.7     | 100.0        |
| POD3      | 94.5     | 100.0        |
| PC        | 90.9     | 99.9         |
| EBD       | 5.3      | 0.1          |

Table 28: PODs for coherence time (ms) and accuray 1.22

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 42.2     | 87.0         |
| POD2      | 48.5     | 71.7         |
| POD3      | 34.6     | 83.3         |
| PC        | 41.8     | 80.7         |
| EBD       | 10.8     | 0.3          |

Table 29: PODs for ground layer fraction (dimensionless) and accuray 0.0

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 74.0     | 99.1         |
| POD2      | 91.2     | 99.1         |
| POD3      | 75.1     | 98.2         |
| PC        | 83.2     | 99.0         |
| EBD       | 12.3     | 0.5          |

Table 30: PODs for ground layer fraction (dimensionless) and accuray 0.14

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## 2 Title for Day 1

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## 2.1 Title for Figures of Day 1

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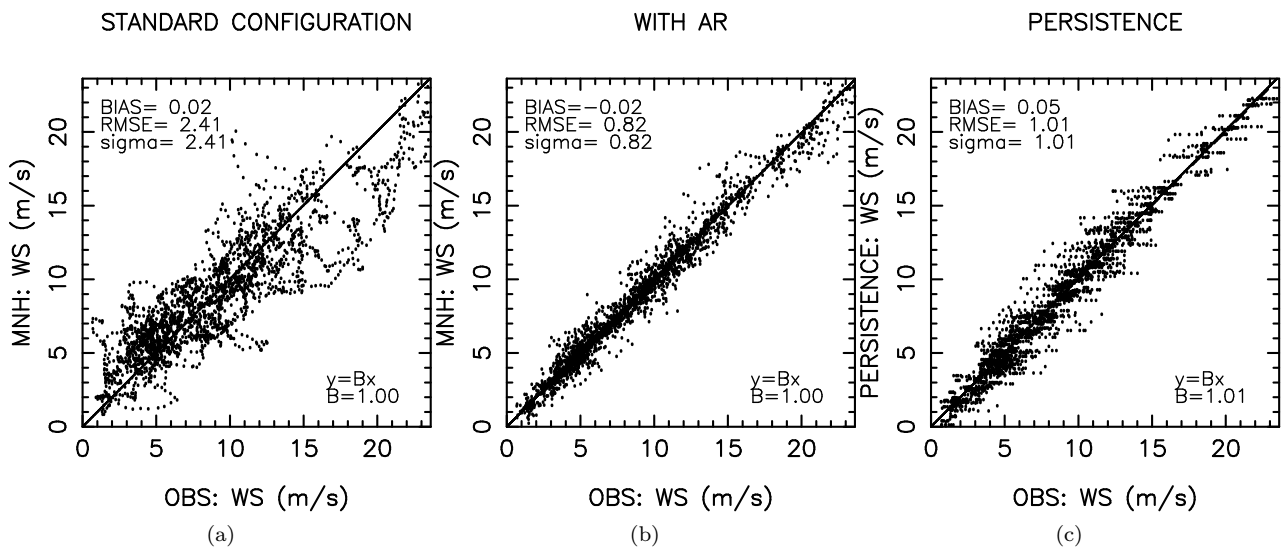


Figure 8: wind speed ( $ms^{-1}$ ): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

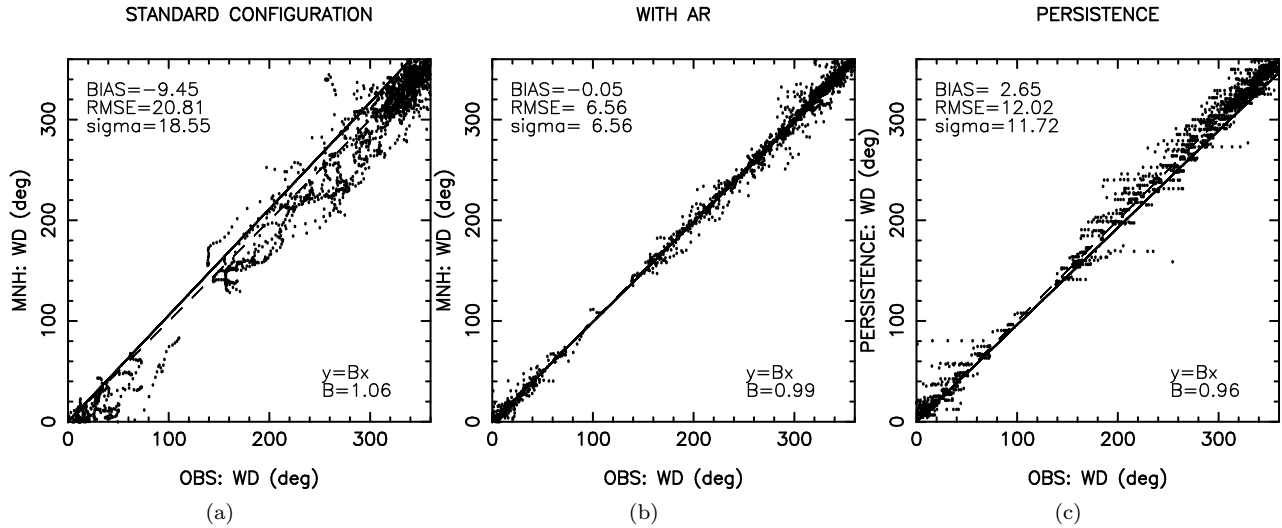


Figure 9: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

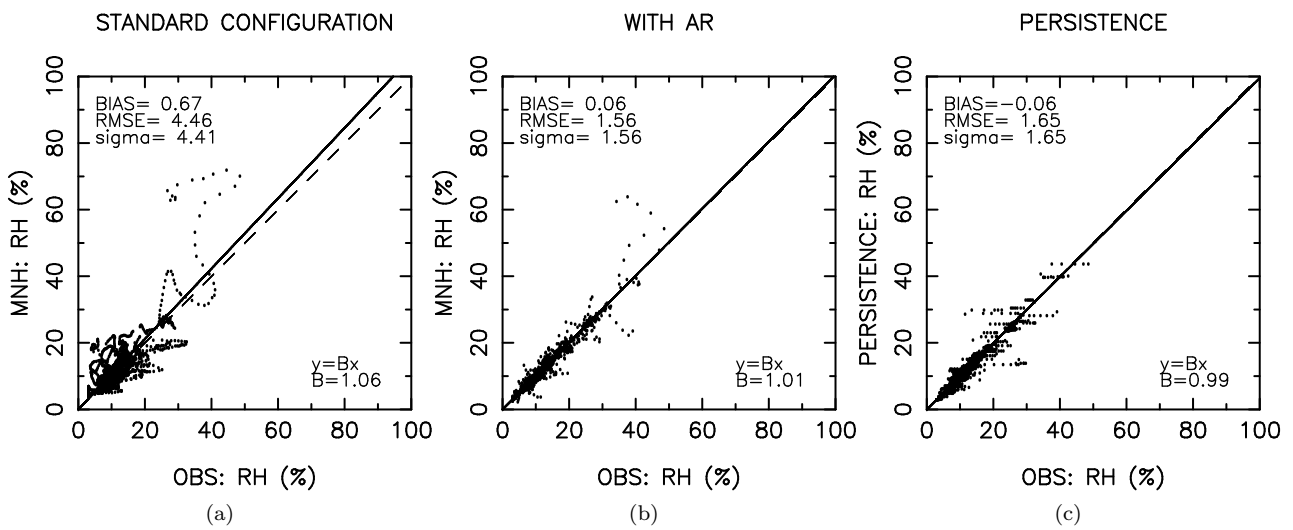


Figure 10: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

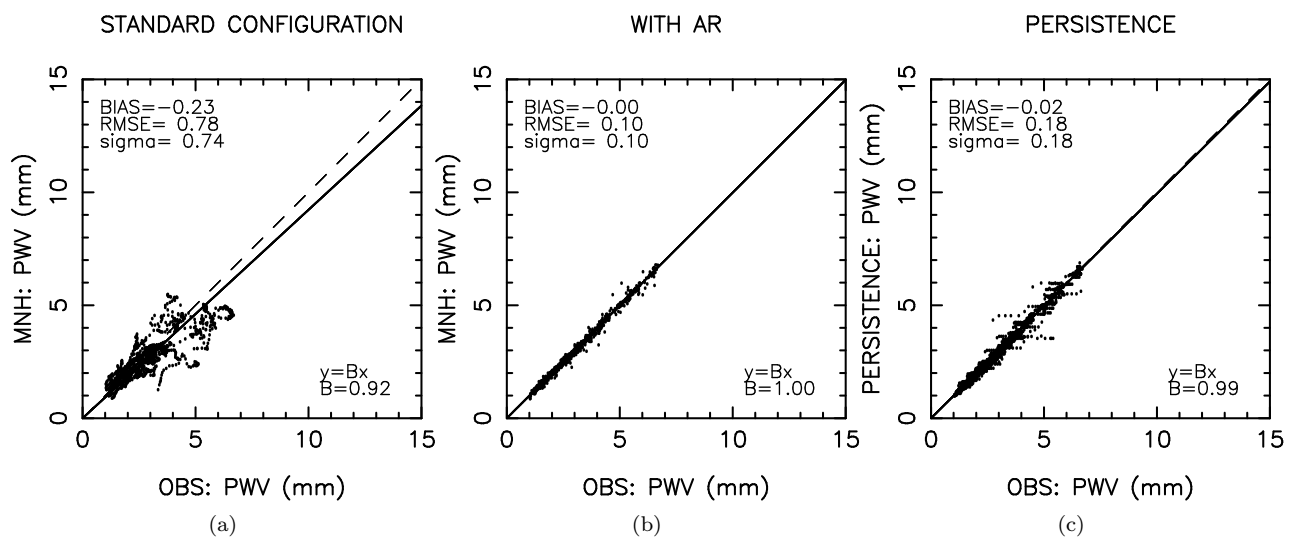


Figure 11: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

## 2.2 Title for Statistics of Day 1

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### 2.3 Title for Contingency tables of Day 1

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| wind speed<br>$ms^{-1}$                                                   | Observations    |                 |           |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                           | ws < 5.7        | 5.7 < ws < 10.5 | ws > 10.5 |
| Model data                                                                | ws < 5.7        | 430             | 63        |
|                                                                           | 5.7 < ws < 10.5 | 370             | 638       |
|                                                                           | ws > 10.5       | 0               | 98        |
| Sample size: 3996; PC=70.5%; EBD=0.3%; POD1=53.8%; POD2=79.8%; POD3=77.8% |                 |                 |           |

Table 31: Contingency table for wind speed ( $ms^{-1}$ ) in standard configuration

| wind direction<br>degree                                                              | Observations |      |       |      |
|---------------------------------------------------------------------------------------|--------------|------|-------|------|
|                                                                                       | North        | East | South | West |
| Model data                                                                            | North        | 1202 | 38    | 0    |
|                                                                                       | East         | 1    | 42    | 8    |
|                                                                                       | South        | 0    | 0     | 251  |
|                                                                                       | West         | 196  | 0     | 10   |
| Sample size: 3996; PC=83.4%; EBD=0.0%; POD1=85.9%; POD2=52.5%; POD3=93.3%; POD4=75.6% |              |      |       |      |

Table 32: Contingency table for wind direction (degree) in standard configuration

| relative humidity<br>percent                                              | Observations    |                 |           |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                           | rh < 8.9        | 8.9 < rh < 12.0 | rh > 12.0 |
| Model data                                                                | rh < 8.9        | 479             | 255       |
|                                                                           | 8.9 < rh < 12.0 | 223             | 333       |
|                                                                           | rh > 12.0       | 97              | 210       |
| Sample size: 3996; PC=59.1%; EBD=4.5%; POD1=59.9%; POD2=41.7%; POD3=75.7% |                 |                 |           |

Table 33: Contingency table for variable relative humidity (percent) in standard configuration

| precipitable water vapor<br>mm                                            | Observations    |                 |           |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------|
|                                                                           | pwv < 2.1       | 2.1 < pwv < 3.2 | pwv > 3.2 |
| Model data                                                                | pwv < 2.1       | 330             | 61        |
|                                                                           | 2.1 < pwv < 3.2 | 51              | 278       |
|                                                                           | pwv > 3.2       | 0               | 42        |
| Sample size: 1998; PC=76.4%; EBD=0.7%; POD1=86.6%; POD2=73.0%; POD3=69.5% |                 |                 |           |

Table 34: Contingency table for variable precipitable water vapor (mm) in standard configuration

## 2.4 Title for PODs tables of Day 1

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| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 53.8     | 91.9         |
| POD2      | 79.8     | 90.4         |
| POD3      | 77.8     | 93.0         |
| PC        | 70.5     | 91.7         |
| EBD       | 0.3      | 0.0          |

Table 35: PODs for wind speed ( $ms^{-1}$ )

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 85.9     | 96.2         |
| POD2      | 52.5     | 92.5         |
| POD3      | 93.3     | 97.8         |
| POD4      | 75.6     | 91.1         |
| PC        | 83.4     | 95.2         |
| EBD       | 0.0      | 0.0          |

Table 36: PODs for wind direction (degree)

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| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 59.9     | 91.0         |
| POD2      | 41.7     | 88.8         |
| POD3      | 75.7     | 95.5         |
| PC        | 59.1     | 91.8         |
| EBD       | 4.5      | 0.0          |

Table 37: PODs for relative humidity (percent)

| PARAMETER | STANDARD | WITH AR (1H) |
|-----------|----------|--------------|
| POD1      | 86.6     | 96.3         |
| POD2      | 73.0     | 95.3         |
| POD3      | 69.5     | 96.6         |
| PC        | 76.4     | 96.1         |
| EBD       | 0.7      | 0.0          |

Table 38: PODs for precipitable water vapor (mm)